

GrabIt

AI-Powered Interview Preparation & Career Readiness Platform

A full-stack, multi-modal AI system combining generative-AI question intelligence, real-time computer-vision behavioral analysis, speech-based simulation, and a custom-built neural network for candidate evaluation.

Major Project Documentation

Prepared for Academic / College Major Project Submission

1. Project Overview

GrabIt is a full-stack, AI-driven interview preparation platform designed to help job seekers practice, evaluate, and improve their interview performance through realistic, AI-simulated mock interviews. The platform goes beyond static question banks by combining **generative AI** (Google Gemini) for adaptive question generation and evaluation, **computer vision** (Google MediaPipe) for real-time behavioral and eye-contact analysis, **speech recognition and synthesis** for a natural, voice-driven interview experience, and a **custom-built neural network** (implemented from first principles, without any ML framework) that fuses multiple performance signals into a final hiring-readiness verdict.

The application is built as a modern single-page React application backed by Supabase (PostgreSQL, Authentication, and Object Storage), and is designed to be a realistic, end-to-end demonstration of applying AI across the full interview-preparation lifecycle - from resume parsing, to live simulated interviews, to personalized improvement roadmaps.

Project Codename

GrabIt - reflecting the platform's goal of helping candidates "grab" the job opportunity they are preparing for.

2. Problem Statement & Motivation

Job seekers - especially students and early-career professionals - frequently struggle with interview readiness, not because they lack technical knowledge, but because they lack realistic, low-stakes practice environments that give holistic feedback covering *technical accuracy*, *behavioral communication*, *emotional stability*, and *non-verbal body language*. Existing solutions typically address only one dimension:

- Static question banks (e.g. lists of "top 50 interview questions") provide no personalized feedback.
- Text-based mock interview tools ignore body language, eye contact, and vocal delivery entirely.
- Generic AI chatbots are not tailored to a specific job description or role, and don't simulate the live, timed pressure of a real interview.
- Human mock interviews (career counsellors, seniors) are not scalable, are time-constrained, and are not always available on demand.

Why This Problem Statement

This problem was chosen because it sits at the intersection of several technically rich domains - Generative AI, Computer Vision, Speech Processing, and core Machine Learning - while remaining a genuinely useful, demonstrable product with a clear real-world audience (students, job seekers, campus placement cells). It allows a single cohesive system to showcase full-stack engineering, applied AI/ML integration, and an original algorithmic contribution (the custom neural network), rather than being a narrow, single-technique project.

3. Objectives

- Simulate a realistic, role-specific interview experience using AI-generated questions derived from a candidate's resume or a target job description.
- Capture and evaluate spoken responses in real time using speech-to-text, eliminating the friction of typed answers.

- Analyze non-verbal cues (eye contact, facial stability, presence) via webcam using computer vision, without requiring specialized hardware.
- Generate a holistic, multi-dimensional performance report (technical, behavioral, stability, body language) using a large language model.
- Fuse these signals through a custom-built neural network to produce a single, explainable hiring-readiness verdict.
- Maintain a persistent history of interview sessions and generate a personalized improvement roadmap based on recurring weak areas.
- Provide supporting tools - a curated question library, an AI career-mentor chatbot, resume/JD parsing, and technology learning roadmaps - to form a complete interview-preparation ecosystem.

4. Technology Stack

4.1 Frontend

Technology	Purpose
React 19	Core UI library, component-based architecture
Vite	Build tool and development server
Framer Motion	Page transitions and micro-interactions/animations
Lucide React	Icon system
Vanilla CSS (custom design system)	Styling, theming, layout

4.2 Backend & Cloud Services

Technology	Purpose
Supabase (PostgreSQL)	Primary database - interviews, chat sessions, roadmaps, documents
Supabase Auth	User authentication and session management
Supabase Storage	Cloud storage for recorded interview videos
Supabase Edge Functions	Serverless function for automated email report dispatch

4.3 Artificial Intelligence / Machine Learning

Component	Technology / Approach
Question generation & evaluation	Google Gemini API (multi-model fallback: Gemini 1.5 / 2.0 Flash & Pro)
Resume / JD understanding	LLM prompt-based role & question extraction with JSON-schema parsing
Facial & behavioral analysis	Google MediaPipe FaceLandmarker (52 facial blendshapes, real-time)
Speech-to-Text	Web Speech API (SpeechRecognition), continuous, interim + final results
Text-to-Speech	Web Speech Synthesis API (natural voice selection)
Candidate scoring model	Custom Multi-Layer Perceptron (MLP) - built from scratch in JavaScript with forward propagation & backpropagation (no ML framework used)

4.4 Document Processing

Technology	Purpose
pdfjs-dist	Client-side PDF text extraction (resume/JD parsing)
mammoth.js	DOCX text extraction

4.5 Tooling

- ESLint - code quality and consistency
- Environment-based configuration (.env.local) for API keys (Gemini, Supabase)

5. System Architecture

GrabIt follows a client-heavy architecture: most AI orchestration (prompt construction, response parsing, real-time video inference) runs directly in the browser, while Supabase provides authentication, persistence, and storage as a managed backend-as-a-service. This keeps the system simple to deploy while still supporting multi-user, cloud-synced data.

5.1 High-Level Flow

- **Landing & Auth** - Unauthenticated users see a marketing landing page, then sign up/log in via Supabase Auth.
- **Resume/JD Ingestion** - User pastes a job description or uploads a PDF/DOCX resume, parsed client-side.
- **Question Generation** - Extracted text is sent to Gemini, which returns a detected role and a tailored question set (JSON).
- **Live Interview Simulation** - Questions are read aloud (TTS); the candidate answers via microphone (STT) while the webcam feed is analyzed frame-by-frame (MediaPipe) for eye contact and stability; audio/video is recorded locally via MediaRecorder.
- **AI Evaluation** - On completion, the full transcript and behavioral score are sent to Gemini, which returns technical, behavioral, stability, and body-language scores plus a qualitative report.
- **Neural Verdict** - The four normalized scores are passed into the custom-built neural network, which outputs a single hiring-readiness percentage and verdict label.
- **Persistence** - The session (scores, report, verdict, recorded video) is saved to local storage immediately, then synced to Supabase in the background; a summary email is dispatched via a serverless Edge Function.
- **Feedback Loop** - Historical sessions feed the Dashboard, History, and Improvement Roadmap modules, closing the loop with targeted practice recommendations.

5.2 Data Model (Supabase Tables)

Table	Purpose
interviews	Stores each interview session's scores, AI report (JSON), and recording URL
chatbot_sessions	Stores AI career-mentor chat conversations per persona
roadmaps	Stores generated improvement roadmaps
user_documents	Stores parsed resume/JD text for reuse

6. Key Features

● AI-Simulated Live Interviews

Voice-driven mock interviews where an AI interviewer ("Alexa") asks role-specific questions aloud and listens to spoken responses in real time, closely mimicking a real interview setting.

● Resume & Job-Description Intelligence

Upload a resume (PDF/DOCX) or paste a job description; the system automatically extracts text, detects the target role, and generates a tailored set of interview questions.

● Real-Time Computer-Vision Behavioral Tracking

Uses on-device facial landmark detection (MediaPipe) to continuously estimate eye contact, emotional stability, and confidence from webcam video, with a live visual tracking overlay.

● AI-Generated Performance Reports

After each session, Gemini analyzes the full conversation transcript to produce scores across four dimensions (technical, behavioral, stability, body language) along with a written summary, strengths, weaknesses, and actionable next steps.

● Custom Neural Network Hiring Verdict

A hand-built multi-layer perceptron (input to hidden to output, sigmoid activation, backpropagation) fuses the four performance scores into a single, explainable hiring verdict - implemented without relying on any external ML library.

● Session Recording & Cloud Sync

Interview audio/video is recorded locally and uploaded to Supabase Storage; all session data is synced to the cloud with a local-storage fallback for offline resilience.

● Personalized Improvement Roadmap

Aggregates patterns across past interview sessions to recommend specific topics and drills the candidate should focus on next.

● Curated Practice Question Library

A searchable, filterable bank of technical, behavioral, system-design, and HR questions for focused, single-question practice.

● AI Career Mentor Chatbot

A multi-persona conversational assistant (Career Mentor, Tech Tutor, Mock Interviewer) with persistent, cloud-synced chat history.

● Technology Roadmaps

Structured learning paths across Frontend, Backend, AI & Data, and DevOps & Cloud stacks, each linkable directly into a targeted practice drill.

● Session History & Analytics Dashboard

A dashboard summarizing past performance trends, scores over time, and quick access to start new practice sessions.

● Opportunities & Billing Modules

Illustrative modules for AI-matched job opportunities and a tiered subscription plan (Starter / Professional / Enterprise), demonstrating the platform's product/monetization design - implemented as UI-complete modules for demonstration.

7. Custom Neural Network - Technical Deep Dive

To demonstrate applied understanding of core machine learning principles (rather than relying solely on third-party APIs), GrabIt includes a neural network implemented entirely from scratch in JavaScript - no TensorFlow.js, PyTorch, or scikit-learn.

Parameter	Value
Architecture	4 input neurons -> 6 hidden neurons -> 1 output neuron
Inputs	Technical Score, Behavioral Score, Stability Score, Body-Language Score (normalized 0-1)
Output	Hire Probability (0-1)
Activation Function	Sigmoid
Training Algorithm	Backpropagation with gradient descent
Learning Rate	0.5
Training Regime	Trained on labelled synthetic candidate profiles for 2000 epochs on each session start

Design note: the network is currently trained on a small, hand-curated synthetic dataset representing archetypal candidate profiles (e.g. "strong technically but weak behaviorally", "well-rounded average", "high presence, average substance"). This is sufficient to demonstrate a working forward/backward-propagation pipeline end-to-end; for a production system this would be extended with a larger, real-world labelled dataset - a natural direction for future work (see Section 10).

8. Use Cases

Actor	Use Case
Final-year / graduating student	Practice placement-season interviews with role-specific, AI-generated questions before campus recruitment drives.
Job-switching professional	Simulate interviews tailored to a specific job description before applying/interviewing.
College placement cell / TPO	Provide a scalable, on-demand mock-interview tool to an entire batch without needing one-on-one human mock interviewers.
Non-native English speaker	Practice verbal fluency and confidence in a low-stakes, judgment-free AI environment.
Self-improvement / repeat learner	Track progress over multiple sessions via the History dashboard and follow a personalized improvement roadmap.

9. What Makes This Project Stand Out

- **Multi-modal AI fusion** - combines LLM reasoning, computer vision, and a hand-built neural network in a single coherent pipeline, rather than a single-technique demo.
- **Genuine algorithmic contribution** - the scoring neural network is implemented from first principles (forward pass + backpropagation), not just an API call, satisfying evaluators who look for original technical depth.
- **Realistic, voice-first UX** - full speech-in / speech-out interview flow, closer to a real interview than typical text-based mock interview tools.
- **End-to-end product thinking** - spans acquisition (landing page), the core AI experience, retention (history, roadmap), and monetization (billing tiers), reflecting real product/engineering scope.
- **Resilience engineering** - Gemini API calls use a cascading model/version fallback strategy with local-storage-first persistence, showing awareness of real-world reliability concerns.
- **No specialized hardware required** - all computer vision and speech processing run in-browser using the user's existing webcam and microphone.

10. Current Limitations & Future Scope

10.1 Known Limitations

- The custom neural network is trained on a small synthetic dataset and is illustrative rather than production-grade.
- Live in-session analytics (word/keyword-based scoring) are heuristic; the authoritative evaluation happens post-session via the LLM.
- Speech recognition depends on the Web Speech API, which is best supported in Chromium-based browsers.
- Opportunities and Billing modules currently use static/sample data to demonstrate the intended product experience.

10.2 Future Scope

- Expand the neural network's training data using real, anonymized/labelled interview outcome data for improved accuracy.
- Add support for additional languages (multilingual STT/TTS) to widen accessibility.
- Integrate a real job-matching API to power the Opportunities module with live listings.
- Add peer-to-peer / human mock-interview scheduling as a complement to AI sessions.
- Introduce payment gateway integration to make the Billing module fully functional.
- Add a proctoring-style integrity layer (tab-switch/attention detection) for formal assessment use cases.

11. Conclusion

Grablt demonstrates a complete, technically diverse application of AI to a real, relatable problem - interview readiness. By combining generative AI, computer vision, speech interfaces, and an original machine learning model within a single full-stack product, it offers strong depth across multiple evaluation criteria: technical breadth, originality, real-world usefulness, and product completeness. It is well suited for submission as a college major project, with clear, well-scoped avenues for future enhancement.